

Tong (Stone) Shi

RESEARCH CV

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RESEARCH INTERESTS

Human-centered AI; urban transportation resilience; multimodal accessibility; spatial equity; environmental exposure; geospatial analysis.

EDUCATION

University of California, Berkeley

M.S., Civil and Environmental Engineering (Transportation Engineering)
Aug 2023 - May 2024 | GPA: 3.2/4.0

University of California, Santa Barbara

B.A., Geographical Information Science
Minor: Architecture and Urban History
Aug 2019 - Jun 2022 | GPA: 3.9/4.0

APPLIED AI AND PRODUCT EXPERIENCE

Ditto

Product Manager, Prompt Engineer | Feb 2025 - present

- Developed a GIS-driven location-recommendation system using Python, GeoPandas, GIS tools, and routing APIs to incorporate mobility patterns, public-transit access, and accessibility zones.
- Designed preference-exclusion logic and prompts to connect user constraints with location recommendations and scenario-based decision support.

Emuser

Product Strategy and User Experience | Project Contributions | 2025 - 2026

- Helped define an experience-first career agent that connects a reusable record of a user's projects, skills, and experiences to job-specific recommendations and application materials.
- Contributed to web-app and browser-extension workflow design, including Auto, Hybrid, and Manual modes, profile-completion checks, progress feedback, and manual job-description entry when extraction fails.
- Developed product positioning around contextual matching and user agency; translated the team's concept into pitch narratives and interaction proposals for early-career applicants and career changers.
- Contributed to product discussions on transparent matching criteria and synchronized profile information, identifying evaluation and data-consistency requirements for trustworthy AI assistance.

Early-stage, mission-driven career-support initiative developing AI-assisted profile building and application workflows.

RESEARCH EXPERIENCE

Socioeconomic Exposure to California Airport Emissions

Project Leader | Institute of Transportation Studies, UC Berkeley | Jan 2024 - Oct 2025

Advisor: Dr. Jasenka Rakas

- Analyzed the spatiotemporal distribution of emissions around 66 California airports and the socioeconomic characteristics of nearby communities.
- Integrated spatial overlays, EPA data, and CalEnviroScreen indicators to examine environmental exposure and inform airport sustainability and mitigation questions.

Spatial Distribution of Accidents in Airport Safety Zones

Research Assistant | Caltrans and NEXTOR III | Jan 2024 - Jul 2024

Advisors: Dr. Jasenka Rakas and Dr. Sarah Lindbergh

- Analyzed accident locations across 259 California airports using NTSB records and digitized runway and safety-zone geometries.
- Applied relative-distance calculations and point-pattern analysis to examine clustering and provide evidence relevant to safety-zone dimensions and risk assessment.

Urban Perception in Biking Mobility: Bogota

Project Leader | HuMNet Lab, UC Berkeley | Sep 2023 - Apr 2024

Advisor: Dr. Marta Gonzalez

- Used Google Street View imagery and PyTorch-based perception modeling to study urban environments encountered during daily mobility.
- Led development of citywide perception layers spanning more than 90,000 road segments; presented related research at the 2024 AAG Annual Meeting.

Milpa Cycle and Labor Density at Tikal

Laboratory Manager | MesoAmerican Research Center, UCSB | Oct 2021 - Apr 2023

Advisor: Dr. Anabel Ford

- Supervised a nine-person GIS team digitizing hydrological networks and verified more than 300 well locations using archaeological records and elevation data.
- Developed watershed and water-supply models for settlement and cultivation analysis; presented at SAA and AAG in 2023 and received second place in the AAG GISS Student Honor Paper Competition.

Streamflow Recurrence and Flood Likelihood

Undergraduate Research Assistant | Hydro Research Group, UCSB | Nov 2020 - Apr 2021

Advisors: Dr. Keith Clarke and Mike Johnson

- Compiled a recurrence-interval dataset of 169,776 observations from USGS monitoring records and developed workflows to analyze streamflow and flood likelihood.
- Communicated time-series analyses through RShiny, Tableau, and RMarkdown visualizations.

SCHOLARLY OUTPUTS AND ADDITIONAL EXPERIENCE

MANUSCRIPT IN PREPARATION

Kondo, G., **Shi, S.**, Anfaresi, N., and Rakas, J. Assessing Flight Network Dynamics and Emission Impacts of California's Intra-State Aviation: A Study on Environmental Justice and Air Traffic Network. Manuscript in preparation; listed as preparing for submission in the 2025 CV.

TECHNICAL RESEARCH OUTPUT

Lindbergh, S., **Shi, S.**, and Rakas, J. (2024). Spatial Distribution Analysis of Aircraft Accident in Airport Safety Zones. FAA NEXTOR III, UC Berkeley.

CONFERENCE PRESENTATIONS

Shi, S., Wang, R., and Gonzalez, M. (2024). How does my cognition change on the roadway? Using street view imageries to assess urban perception in human daily mobility. AAG Annual Meeting.

Shi, S., Kresse, M., Moran, T., Ford, A., and Carr, R. (2023). Modeling the Milpa at Tikal: New dimensions of the Carr and Hazard map. 88th SAA Annual Meeting.

Shi, S., Ramirez, A., and Ford, A. (2023). Visualize Sustainable Land Use in the Tropical Maya Forest: Revitalizing the Old Tikal Map for New Geospatial Analyses. AAG Annual Meeting.

ADDITIONAL PRODUCT EXPERIENCE

TossTo.ai

Founder, Product Manager | Nov 2023 - Jan 2025

- Led prototype design and beta feedback cycles for an LLM-assisted travel and dining product; used input from more than 30 primary users to inform the concept and point-of-interest recommendations.

YoMio BEYOUNG Co., Ltd.

Co-Founder, Product Manager | Jan 2024 - present

- Led a four-person team developing an AI character-interaction product and contributed retrieval-augmented generation design grounded in brand knowledge and user preferences.

SELECTED HONORS

- Departmental Fellowship, UC Berkeley, 2024 (\$25,000).
- Student Opportunity Fund, UC Berkeley, 2023 (\$1,350).
- Second Place, AAG GISS Student Honor Paper Competition, 2023.
- Academic Excellence Award; Outstanding Achievement in Geography Major; High Honors, UCSB, 2022.
- UCSB Summer Creative Computing Initiative Scholarship, 2021; Gene and Susan Lucas Undergraduate Research Fund, 2020.